Parallel Pipeline for Genomic Selection

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1 Introduction

The ppGS library was developed for easy and parallel computation of a pipeline for genomic selection (GS) on potato color traits from AGROSAVIA's Colombian Central Collection (CCC).

2 Installation

The package **ppGS** is sitting on GitHub repo: `https://github.com/luisgarreta/ppGS`. To install it, first, you need to install the **devtools** package. You can do this from CRAN. Invoke R and then type:

```
install.packages("devtools")
```

Load the devtools package.

```
library(devtools)
```

In most cases, you just use install_github("author/package"). For the package **ppGS**, you'd type:

```
install_github("luisgarreta/ppGS")
```

3 Genotypic data

The CCC genotypic data includes two tables: a genotype matrix and a color phenotype table. The library has a utility function called **gs_datasets** that creates different datasets. For example, to get a copy of the original genotype/phenotype, call the function as follows.

```
gs_datasets (type="sources")
```

This function extracts the genotypic data (genotype and phenotype), creates a "sources" directory and copies them as files into it.

3.1 Genotype structure

The genotype consists of a table with markers in the rows, observations in the columns and tetraploid genotype values in each cell. The following is an excerpt from this file:

```
geno = read.csv ("sources/sources-genotypes-CCC-Andigena-ClusterCall2020.csv",
                 check.names=F)
geno [1:5, 1:8]

Markers  15060006  15060013  15060016  15060021  15060025  15060027  15060039
c1_1         4         4         4         4         4         4         3
c1_1000      2         1         2         2         2         2         1
c1_10000     1         0         1         1         0         0         1
c1_10001     3         4         3         3         4         4         3
c1_10011     1         0         1         1         0         0         2
```

The original marker names were simplified from "solcap_snp_XXX" to "XXX" (e.g., "solcap_snp_c1_1" has been changed to "c1_1"). And the names of the observations were changed to their record number, as opposed to their old access identifier (e.g., "And_10" has been changed to "15060006").

3.2 Phenotype structure

The phenotype data consists of a table with the observations in the rows, the color phenotypes in the columns, and the phenotype values in each cell. An excerpt from this file is shown below:

```
phenos= read.csv ("sources/sources-phenotypes-CCC-Andigena-ColorTraitsHCL.csv",
                  check.names=F)
phenos [1:5, 1:6]

Samples  Tallo.LCH.L  Tallo.LCH.C  Tallo.LCH.H  PriFlor.LCH.L  PriFlor.LCH.C
15060006      28           16          359           52             45
15060013      NA           NA           NA           54             49
15061904      28           16          359           36             55
15060016      28           16          359           45             57
15060021      28           16          359           37             43
```

Each row corresponds to an observation and each column to a color phenotype, in total there are 24 phenotypes corresponding to eight color characteristics measured in potato and decomposed into three components of the HCL color space with "H" for huge, "C" for chroma and "L" for luminance.

The eight characteristics are "PriFlower" for flower primary color, "Baya" for berry color, "PriTuber" for tuber primary color, "SecTuber" for tuber secondary color, "Pulpa" for flesh color, "PriBrote" for bud primary color, and "SecBrote" for bud secondary color.

4 Genomic selection for multiple phenotypes

The gs_multi function performs the genomic selection process for multiple phenotypes. The function requires two arguments: the name of the configuration file (parameters in YAML format) and the name of the output directory for the results. The configuration file contains the following parameters:

Table 1: Configuration parameters for GS with multiple phenotypes.

PARAMETER	**DESCRIPTION**
trainingGenoFile	Path name of the genotype matrix with the observations to be used for model training.
trainingPhenoFile	Path name of the phenotype table with the observations to be used for model training.
targetGenoFile	Path name of the genotype matrix with the observations that will be used to calculate the GEBVs.
targetPhenoFile	(Optional) Path name of the actual values of the observations used to calculate the GEBVs.
markersFile	(Optional) Path name of the table with the names of the selected markers (SNP column) for training/testing the models.
nMarkers	Number of markers to be taken from the selected markers file (markersFile).
methods	List of genomic breeding prediction methods. The available methods are: BA BB BC BL BRR GBLUP EGBLUP RKHS
nTimes	Number of independent replicates for the cross-validation.
nFolds	Number of folds for the cross-validation.
nCores	Number of processor cores to use for parallel processing.

4.1 Training/testing datasets

The parameters for running GS with the **gs_multi** function need training and target datasets for genotype and phenotype. Just to test this function, you can generate these datasets using the **gs_datasets** function in this way:

```
gs_datasets (type="small")

>>> Copying small datasets to /home/lg/agrosavia/COLOR-TRAITS/GS-PKG/ppGS/vignettes
small-config.yml
small-target-geno.csv
small-target-pheno.csv
small-training-geno.csv
small-training-pheno.csv
```

The above call using type="small" generates a small dataset containing 1000 markers and 6 phenotypes. In contrast, you can generate a complete dataset containing 4640 markers and 24 traits in this way:

```
gs_datasets (type="full")
```

4.2 Running GS

Running GS can take a few minutes or several minutes depending on your machine, the number of cores used in parallel, and the type of data sets. The small dataset may take a few minutes (1 to 3 minutes, with 4 cores), but the full dataset may take several minutes (8 to 10 minutes, with 4 cores).

To run GS with the small datasets, execute:

```
gs_datasets (type="small")
setwd ("small")
gs_multi (configFile="small-config.yml", outputDir="smallGS")
```

Now, to run GS with the full datasets, execute:

```
gs_datasets (type="full")
setwd ("full")
gs_multi (configFile="full-config.yml", outputDir="fullGS")
```

4.3 Results

The **gs_multi** function generates two types of results: first, a table with the GEBVs for each phenotype, and second, boxplots comparing the predictions of the phenotypes. In both outputs, the values are calculated with the best prediction method for each phenotype, i.e., the methods with maximum predictive ability (PA), where PA is measured by the correlation between the actual phenotype and the GEBV of the cross-validation.

4.3.1 GEBVs for each phenotype

GEBVs are given for each observation (rows) and each phenotype (columns). If the target phenotype file with actual values was specified in the configuration file, these values are shown next to the GEVBs. An excerpt of this table is shown below:

Table 2: GEBVs for phenotypes. Observations in rows, actual value and GEBVs in columns.

SAMPLES	Baya.LCH.C	Baya.LCH.C.GEBV	Baya.LCH.H	Baya.LCH.H.GEBV	Baya.LCH.L	Baya.LCH.L.GEBV
15061942	45.89	48	117.044	118	59.728	66
15061763	44.876	38	116.341	132	59.183	53
15061801	38.915	48	116.67	118	56.073	66
15061494	44.931	44	117.898	117	59.262	54
15061613	43.561	45	116.306	128	57.094	58
15062137	45.401	44	117.458	117	59.109	54
15061319	46.257	45	116.11	115	59.88	72

4.3.2 Best prediction methods for phenotypes

This result is provided in two ways: a graphical comparison of the PA for each phenotype calculated for the best method, and a table with the PA results from cross-validation. The figure below shows the GS results using complete datasets.

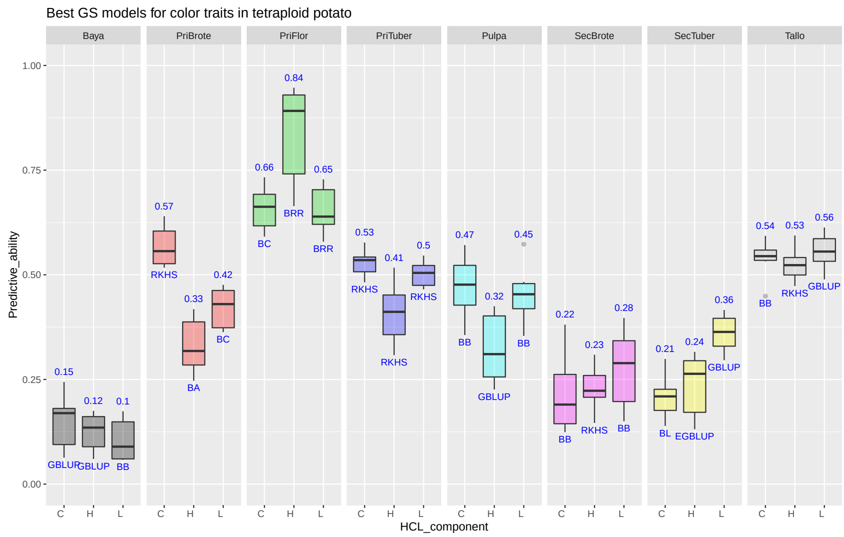

Figure 1: Best prediction methods using full markers.

4.4 Results of GS using selected markers

The configuration file for GS contains two optional parameters called "markersFile" and "nMarkers", the first for a file name of a table with selected marker names (e.g. best markers resulting from a GWAS analysis) and the second for the number of markers to take from this table. If these parameters are present (not NULL), the **gs_multi** function takes only these markers to execute the whole GS process. The following figure shows the results of a GS using the results of a previous GWAS analysis and selecting from it the best 50 markers:

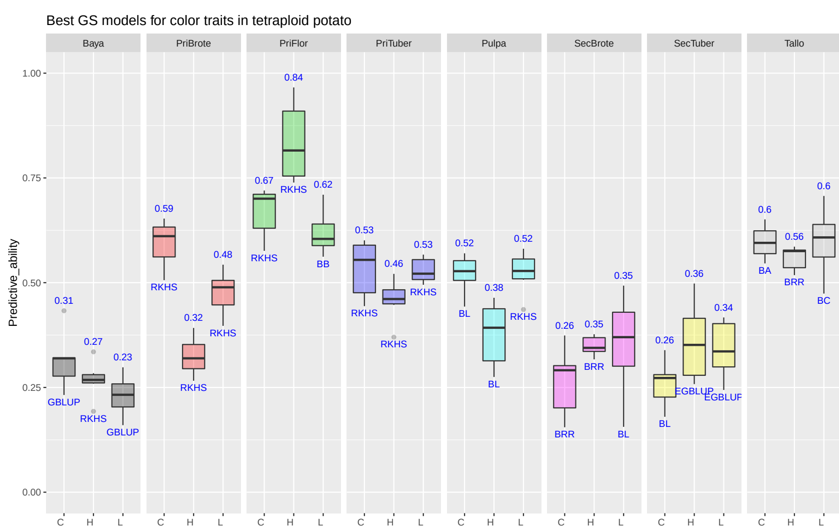

Figure 2: Comparison of prediction methods using 50 selected markers from GWAS.

To compare the above results using selected GWAS markers, the following figure shows the results of GS using 50 random markers from the full set of markers:

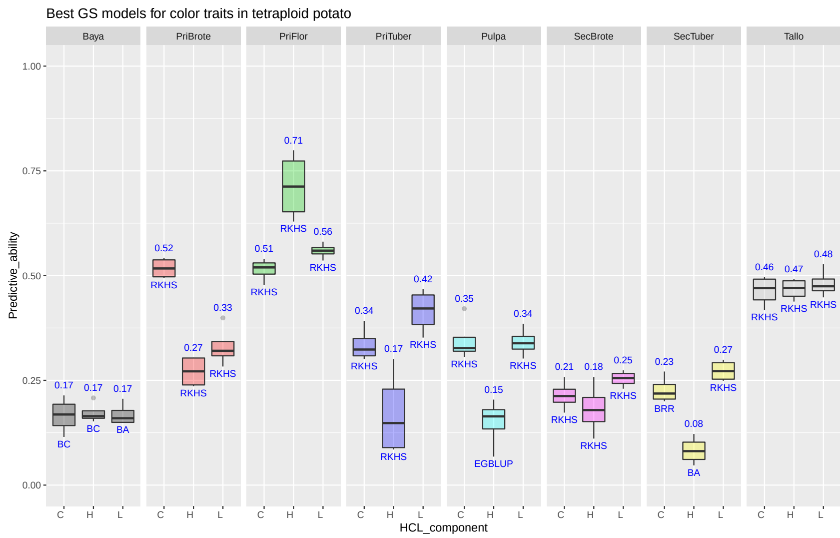

Figure 3: Comparison of prediction methods using 50 random markers.

5 Genomic selection for single phenotype

To run the GS pipeline for a single phenotype, the package has the **gs_single** function that uses two parameters: a configuration file name with the GS parameters (YAML format) and an output directory name where the results will be saved.

5.1 Configuration file

The configuration file is similar to the one used in the gs_multi function excluding the parameters "markersFile" and "nMarkers" and including a new parameter "phenoName" which specifies the phenotype name. The following table describes the parameters of this file:

Table 3: Configuration parameters for GS with single phenotypes.

PARAMETER	DESCRIPTION
trainingGenoFile	Filename path for the genotype matrix of observations to be used for training the models.
trainingPhenoFile	Filename path for table with multiple phenotypes of observations to be used for training the models.
targetGenoFile	Filename path for the genotype matrix of observations used for computing the GEBVs.
targetPhenoFile	(Optional) Filename path for actual values of observations used for computing the GEBVs.
phenoName	Name of phenotype to predict GEBVs.
methods	List of genomic breeding prediction methods. The available models are: BA BB BC BL BRR GBLUP EGBLUP RKHS
nTimes	Number of independent replicates for the cross-validation.
nFolds	nCores Number of processor cores to use for parallel processing.

5.2 Running GS

To test the GS run with a single phenotype, you can use the **gs_datasets** function to create training/target genotypes/phenotypes with their respective configuration, change to the directory with the datasets and run **gs_single** as follows:

```
gs_datasets (type="single")
setwd ("single")
gs_single (configFile="single-config.yml", outputDir="singleGS")
```

5.3 Results of GS

The **gs_single** function generates several outputs (Figure 4): (a) Table of GEBVs along with the actual value (if the target phenotype file was specified in the configuration file); (b) Histogram and correlation between GEBVs and training values; (c) Histogram and correlation between GEBVs and the actual values of the phenotypes (if the target phenotype file was specified in the configuration file); (d) Comparison of predictive ability by methods; (e) Run time of the methods; and (f) Correlations between the results of the different methods.

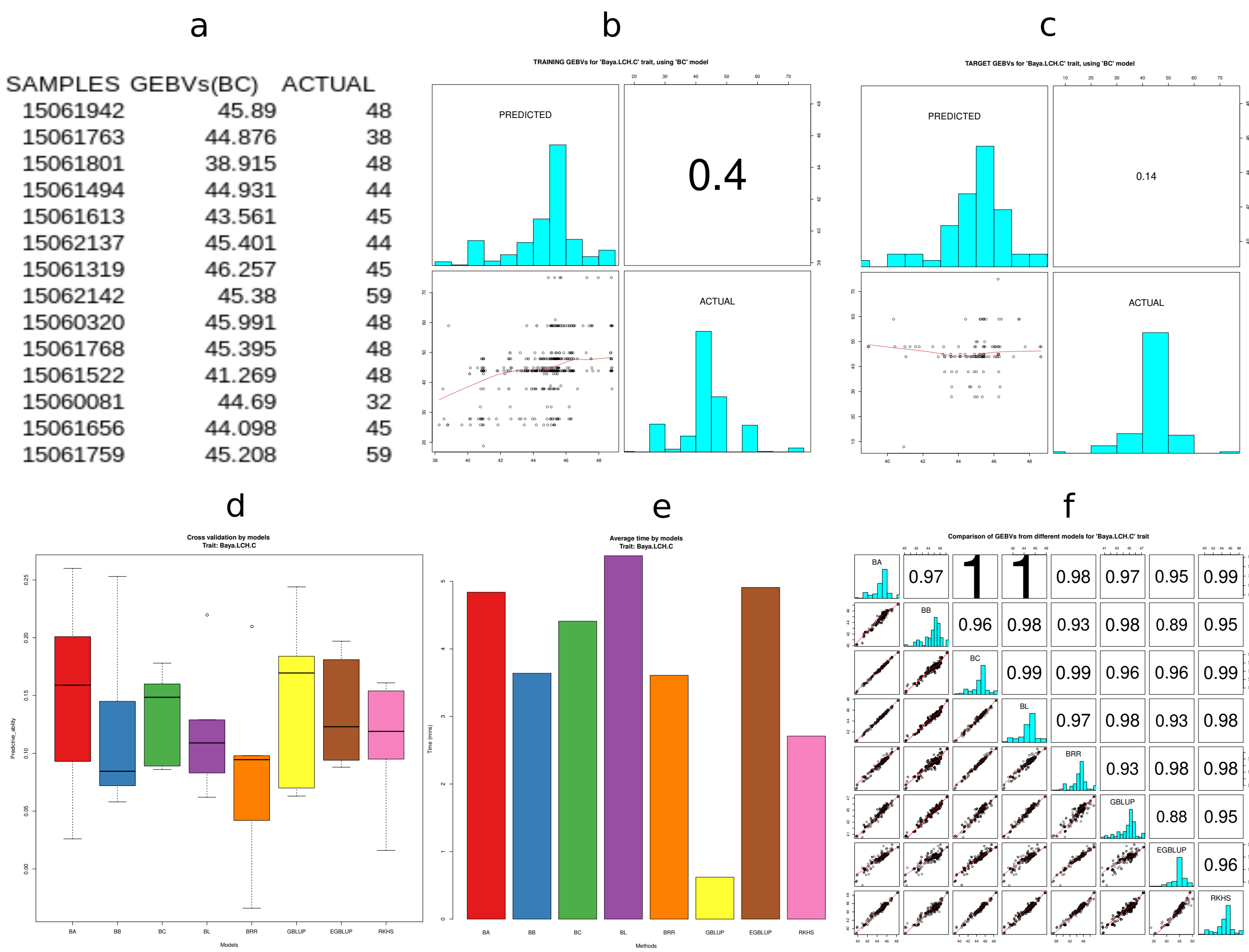

Figure 4: Outputs from GS with single phenotype.

6 Heritability

The heritability for multiple phenotypes can be calculated with the **gs_heritability** function that takes three parameters: genotype matrix filename, phenotype table filename, and output directory name. For example, to calculate the heritabilities of the source genotype and phenotype, call the function as follows:

```
# Get copies of source genotype matrix and phenotypes table
gs_datasets (type="sources")

# Change to "sources" directory
setwd ("sources")

# Calculate heritabilities
gs_heritability ("sources-genotypes-CCC-Andigena-ClusterCall2020.csv",
                 "sources-phenotypes-CCC-Andigena-ColorTraitsHCL.csv",
                 "outputs")
```

Create the "outputs" directory and save in it two files: a table (CSV) with several heritabilities for each phenotype and a summary plot (PDF) comparing the average heritabilities of each phenotype. The summary plot is shown in the figure below:

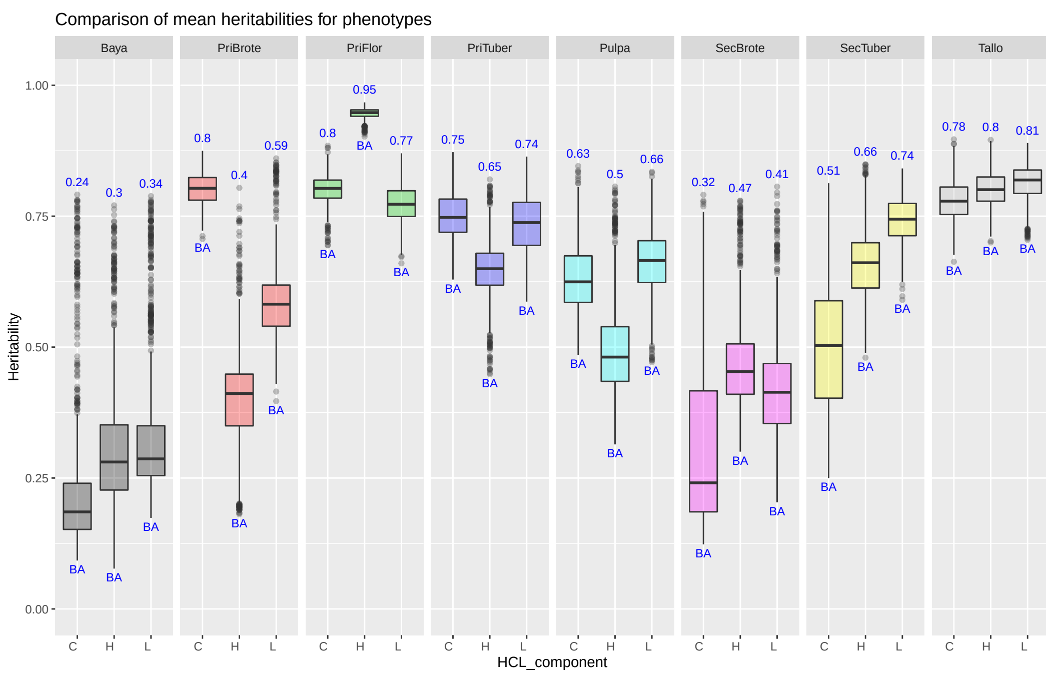

Figure 5: Comparison of mean heritabilities of potato color phenotypes.